

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276319

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Fuller; Kevin S. et al.

SCREENING DEVICE FOR ANALYSIS OF BODILY FLUIDS

Abstract

A screening device for testing saliva for the presence of certain constituents. The device has a D-shaped profile when viewed from above and includes a first wall with planar front face through which all of the test strips are viewed and a curved rear second wall. The device also includes a floor surface with an uppermost portion extending from the rear wall to a lowermost portion at the first wall such that saliva squeezed from a sampler at a location between the upper most portion and the lowermost portion flows downwardly to the test strips.

Inventors: Fuller; Kevin S. (Chaska, MN), Altendorf; Wayne J. (Hickson, ND)

Applicant: Premier Biotech, LLC (Minneapolis, MN)

Family ID: 50681882

Appl. No.: 19/212003

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 16875583 20200515 PENDING child US 19212003

parent US continuation 16008720 20180614 parent-grant-document US 10744507 child US 16875583

parent US continuation 14834534 20150825 ABANDONED child US 16008720

parent US continuation 14079558 20131113 ABANDONED child US 14834534

us-provisional-application US 61725898 20121113

Publication Classification

Int. Cl.: B01L3/00 (20060101); A61B10/00 (20060101); G01N1/18 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 16/875,583 filed May 15, 2020, which is a Continuation of U.S. patent application Ser. No. 16/008,720 filed Jun. 14, 2018, now U.S. Pat. No. 10,744,507, which is a Continuation of U.S. patent application Ser. No. 14/834,534 filed Aug. 25, 2015, which is a Continuation of U.S. patent application Ser. No. 14/079,558 filed Nov. 13, 2013, which claims the benefit of U.S. Provisional Patent Application No. 61/725,898 filed Nov. 13, 2012, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

[0002] This application is directed generally to devices and methods for analysis of the presence of constituents in fluids, and more specifically to devices and methods for analysis of the presence of constituents in bodily fluids for drug test screening.

BACKGROUND OF THE INVENTION

[0003] Various drug screening devices are known for testing a sample of bodily fluids for the presence of certain constituents. These devices typically include a container for receiving a bodily fluid from a delivery device, and a plurality of reagent test strips arranged to contact the bodily fluid, with each test strip being sensitive to a particular constituent of interest. The targeted constituents for screening can range from controlled substances such as alcohol and marijuana to illegal substances such as cocaine and methamphetamines.

[0004] For screening purposes, it is desirable to test for as many constituents as possible with a single sample, so as to reduce the nuisance factor associated with obtaining multiple bodily fluid samples. This is particularly a problem with saliva where sufficient quantities for multiple tests is problematic. Accordingly, many devices are intended to distribute the sampled bodily fluid among as many test strips as reasonably possible. This is typically accomplished by arranging a plurality of test strips on each of several sides of the container.

[0005] Problems associated with a plurality of test strips arrayed on several sides of a test container include maldistribution of bodily fluid samples to the various test strips and the necessity to inspect more than one side of the container to determine the screening result. A screening device that addresses these maladies would be welcomed.

SUMMARY OF THE INVENTION

[0006] Various embodiments of the invention include a device and method for adequately distributing a sample of bodily fluid among a plurality of test strips and preventing maldistribution of sampled fluid that can lead to an invalid test result, all while displaying the test results on a single face of the screening device. Certain embodiments also enable a slight and passive pressurization of a sampling chamber to augment the wicking or capillary action of the sampled fluid along the test strips.

[0007] One problem with the prior art arrangement of having a plurality of test strips arrayed on several sides of a test container is that the arrangement is prone to operator error. Persons conducting the screening test often hold the screening device in their hand while awaiting the test results. While holding the device, there is a tendency to tilt the proximal face of the device backward for easier reading. Unfortunately, this can cause the bodily fluid under test to pool at the bottom of the distal face of the testing device, thus starving the test strips that are disposed on the

proximal face. The result is an invalid test for the constituents represented by the test strips on the proximal face.

[0008] Structurally, various embodiments of the invention include a container or cup having a first or “viewing” side through which all of the test strips can be viewed. The bottom of the cup is inclined and intersects the viewing side along a junction line that constitutes the lowest portion of the cavity within the cup where the bodily fluid collects. The test strips are oriented so that the detection end proximate the junction line for in contact with the bodily fluid that collects at the low portion of the cavity. By this arrangement, all of the test strips can be inspected by viewing a single surface of the cup, and the inclined surface of the bottom of the cup directs all of the bodily fluid into contact with the detection end of the test strips. Furthermore, the angle of the intersection between the bottom and the viewing side can be steep enough to counter a substantial holding angle induced by the operator.

[0009] In certain embodiments, a three-dimensional gasket is implemented to provide certain sealing and pressurization advantages. The three-dimensional gasket can be made of a soft, pliable material to provide a sealing surface between the gasket and the cup and to provide a seal between a sampling stem that is inserted through the gasket. The pliable material can also cause the three-dimensional gasket to act as a diaphragm; when the sampling stem is inserted into the three-dimensional gasket, the displaced air causes the three-dimensional gasket to expand (as would a diaphragm), thereby causing the air contained by the three-dimensional gasket to become slightly pressurized. The pressurization can assist in the diffusion of the bodily fluid into the test strips, thus augmenting the capillary action between the bodily fluid and the test strip.

[0010] The device can also provide barriers between the detection ends of adjacent test strips to prevent more absorbent or faster absorbing test strips from drawing test fluid away from less absorbent or slower absorbing test strips.

[0011] A feature and advantage of embodiments of the invention is a specimen sampling device for saliva where the saliva is absorbed in a sponge, is squeezed into a saliva pathway of a specimen container, the saliva pathway is pressurized urging the saliva into engagement with a plurality of sample test strips. In embodiments a vent may be provided downstream of where the saliva engages the test strips. In embodiments, the plurality of test strips are each positioned in a plurality of recesses in a row, the recesses having an overflow level whereby fluid that overflows one recess flows into an adjacent recess. In embodiments, the pressurization may be provided by a stem attached to the sponge, the stem having a sliding plunger member. In embodiments the pressurization may be provided by a separate manual squeeze or push pump as part of the specimen sampling device.

[0012] Another feature and advantage of embodiments of the invention is a specimen sampling device for saliva where the saliva is absorbed in a sponge, is squeezed into a saliva pathway of a specimen container, the saliva pathway leading to a plurality of sample test strips aligned in a linear row of at least 8 strips positioned on a substantially planar wall and exposed to be readable through one or more transparent regions, the strips having a sample receiving end that is placed at the bottom of an inclined plate in the saliva pathway. In some such embodiments, embodiments may have a gasket sealingly enclosing a specimen chamber including the inclined plate and a row of recesses into which the at least 8 strips extend. In some such embodiments, a curved, wall is positioned opposite the planar wall.

[0013] An additional feature and advantage of embodiments of the invention is a specimen sampling device for saliva where the saliva is absorbed in a sponge, is squeezed into a saliva pathway of a specimen container, the saliva pathway leading to a plurality of sample test reservoirs, each with test strip extending therein, a gasket with a separate aperture for each respective test strip, the gasket providing sealing between the saliva pathway including and the reservoirs and regions of the container above the saliva pathway thereby preventing unintended migration of the saliva upward.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is an exploded view of a testing assembly for screening of bodily fluids in an embodiment of the invention;

[0015] FIG. 2 is a perspective view of a cup of the testing assembly of FIG. 1 in an embodiment of the invention;

[0016] FIG. 3 is a sectional view of the cup of FIG. 2;

[0017] FIG. 4 is a perspective view of a test strip and sampling stem holder of FIG. 1 in an embodiment of the invention;

[0018] FIG. 5 is a side sectional view of the holder of FIG. 4;

[0019] FIG. 6 is a perspective view of the top of a three-dimensional gasket of FIG. 1 in an embodiment of the invention;

[0020] FIG. 7 is a perspective view of the bottom of the three-dimensional gasket of FIG. 6;

[0021] FIG. 8 is a perspective view of a test strip;

[0022] FIG. 9 is a perspective view of a sampling stem of FIG. 1 in an embodiment of the invention;

[0023] FIG. 10 is a perspective cutaway view of the holder and three-dimensional gasket of FIG. 1 in assembly;

[0024] FIG. 11 is a sectional view of the cup, holder and three-dimensional gasket of FIG. 1 in assembly; and

[0025] FIG. 12 is a perspective cutaway view of the cup, holder and three-dimensional gasket of FIG. 1 in assembly.

DETAILED DESCRIPTION

[0026] Referring to FIG. 1, a testing assembly 20 for screening bodily fluids for the presence of certain constituents is depicted in an embodiment of the invention. The screening device includes a container or cup 22, a test strip and sampling stem holder 24 (hereinafter “holder 24”), a three-dimensional gasket 26 and a sampling stem 28.

[0027] Referring to FIGS. 2 and 3, the cup 22 is depicted in isolation in an embodiment of the invention. The cup 22 includes a plurality of sides 32 and 34 that extend upward from a containment bottom 38 to form a continuous upper edge 36, the sides 32, 34 and containment bottom 38 defining a cavity 42. The sides 32 and 34 can also extend below the containment bottom 38 to establish a base portion 40 with a lower edge 41 registers the testing assembly 20 in an upright position. A first of the sides (side 32 in FIGS. 2 and 3) can be substantially planar. In one embodiment, the containment bottom 38 defines an inclined surface 44 that intersects the plurality of sides 32 and 34. The inclined surface 44 intersects the first side 32 along a junction line 46, defining an acute angle θ between the first side 32 and the inclined surface 44. The junction line 46 is the lowermost extremity of the cavity 42 when the cup 22 is in the upright position.

[0028] While the inclined surface 44 is depicted as being substantially planar, it is noted that the inclined surface 44 is not limited to a planar configuration. For example, the inclined surface could also be convex, with a steeper slope at the junction line 46 than at other locations on the inclined surface. Alternatively, the inclined surface could be concave, with a more gradual slope at the junction line 46 than at other locations.

[0029] In one embodiment, the containment bottom 38 can further include a pedestal portion 48 that extends from the inclined surface 44 into the cavity 42. The pedestal portion 48 can include a top face 49 having a perimeter 50. A securing structure 51, such as a point or barb (as depicted in FIG. 3), can be included on or as part of the top face 49 of the pedestal portion.

[0030] Various embodiments of the cup 22 can also include a syringe access port 53 located proximate a lower corner of the cup 22. The syringe access port comprises a region where the

thickness of the cup **22** is thinned to enable manual puncturing with a syringe.

[0031] The cup **22** can be fabricated from a variety of materials available to the artisan, including but not limited to polycarbonate, styrene-butadiene copolymers (SBCs) such as K-RESIN, and clear styrene.

[0032] Referring to FIGS. **4** and **5**, the holder **24** is depicted in isolation in an embodiment of the invention. The holder **24** includes a cap portion **52** having an access port **54** formed therein. The holder **24** also includes a through passage structure **56** that defines a through passage **58** that passes through the holder **24**. The through passage structure **56** includes a proximal end **60** and a distal end **62**, each being concentric about an insertion axis **64**. The proximal end **60** of the through passage structure **56** includes the access port **54**. The through passage structure **56** can comprise a hollow tube that depends from the cap portion **52** (as depicted), or a bore that extends through a substantially solid holder **24**.

[0033] The holder **24** also includes a planar face **66** that extends downward from the cap portion **52**. The planar face **66** includes a plurality of parallel ribs **68** that define a plurality of elongate channels **72**. The elongate channels **72** each define a channel axis **74**.

[0034] The holder **24** can be fabricated from a variety of materials available to the artisan, including but not limited to polycarbonate, styrene and acrylonitrile butadiene styrene (ABS).

[0035] Referring to FIGS. **6** and **7**, the three-dimensional gasket **26** is depicted in an embodiment of the invention. The three-dimensional gasket **26** includes a passage extension structure **82** that defines an extended passage **84** and a diaphragm portion **86** that depends from and extends radially outward from the passage extension structure **82**. A perimeter structure **88** of the three-dimensional gasket **26** surrounds and depends from the diaphragm portion **86**, the perimeter structure **88** having a lower face **92** that extends below the diaphragm portion **86**. The three-dimensional gasket can also include a notch **91** that, upon assembly, aligns with the syringe access port **53** of the cup, enabling easy access to the chamber **150**. The three-dimensional gasket **26** can also include guide tabs **93** that assist in guiding, registering and seating the three-dimensional gasket into the holder **24** during assembly.

[0036] The perimeter structure **88** includes a front portion **90** having an outer face **94** and an inner face **96**. The outer face **94** is adapted to contact an inner surface of the first side **32** of the cup **22**. In one embodiment, the inner face **96** includes a plurality of riblets **102** that extend inwardly, in the general direction of the extended passage **84**. The riblets **102** can include lower surfaces **104** configured to engage the inclined surface **44** of the containment bottom **38**. In one embodiment, the three-dimensional gasket **26** is fabricated from a soft, pliable material such as polyvinylchloride (PVC) or neoprene.

[0037] Referring to FIG. **8**, a test strip **110** is depicted. The test strip **110** includes a detection end **112** and a non-detection end **113**. The detection end **112** wicks fluid to a sensing zone **114** via capillary action through an absorbent material **116**. The sensing zone **114** is configured with reagents that react and provide a visual indication when a certain constituent is present in the bodily fluid. A width **118** of the test strip **110** is dimensioned to slidably engage a corresponding one of the elongate channels **72**. In the depicted embodiment, the test strip **110** is characterized as having increased thickness at both the detection and non-detection ends **112** and **113** due to the presence of the absorbent material **116**.

[0038] Referring to FIG. **9**, the sampling stem **28** is depicted in an embodiment of the invention. The sampling stem **28** is characterized as having a proximal portion **132** and a distal portion **134**. The proximal portion **132** can include a handle or knob **135** for handling and manipulation of the sampling stem **28**. Certain aspects of the sampling stem **28** are depicted and described at U.S. Patent Application Publication No. 2012/0094303 to Engel et al., which is hereby incorporated by reference herein except for express definitions contained therein. The sampling stem **28** can be fabricated from a variety of materials available to the artisan, including but not limited polycarbonate, styrene-butadiene copolymers (SBCs) such as K-RESIN, and clear styrene.

[0039] In one embodiment, the proximal portion **132** includes structure that is complementary with structure on the cap portion **52** for securing the sampling stem **28** in a fixed position along the insertion axis **64** relative to the cap portion **52**. In the depicted embodiment, the complementary structures comprise a female threaded structure **136** proximate the access port **54** of the cap portion **52** of the holder **24** and a male threaded portion **138** proximate the proximal portion **132** of the sampling stem **28**. Other complementary structures could also be utilized, including a detent-and-groove snap fit structure a bayonet-style lock or a clasp.

[0040] In one embodiment, the sampling stem **28** and the access port **54** are configured to provide a ratchet-and-pawl mechanism for rotationally locking the sampling stem **28** relative to the holder **24**. In the depicted embodiment, the sampling stem **28** includes a pawl **142** distal to the male threaded portion **138**. The pawl **142** is configured to engage ratchet notches **144** (FIG. 4).

[0041] The distal portion **134** of the sampling stem **28** includes a sampling swab **140** that is mounted to a collar **141**. In one embodiment, the collar **141** is comprised of a transparent material to enable ultraviolet activation of bonding agents that are placed between the sampling swab **140** and the collar **141**. The distal portion **134** is configured for insertion through the through passage **58** and extended passage **84**. In one embodiment, the collar **141** is sized to provide a sliding seal between the collar **141** and the extended passage **84**. Also, in one embodiment, the through passage **58** as well as the extended passage **84** is tapered (i.e., has a larger diameter at the proximal end than at the distal end) in order to guide the distal portion **134** without causing it to bind, but still affecting a seal as the collar **141** approaches its final axial location along the insertion axis **64**.

[0042] Referring to FIGS. **10** through **12**, the testing assembly **20** is depicted in various states of assembly. In assembly, the three-dimensional gasket **26** is coupled to the bottom of the holder **24** and the test strips inserted into the respective elongate channels **72** so that the detection end extends into the extended channel. The holder **24** and three-dimensional gasket **26** assembly, with test strips **110** loaded in the channels **72**, is inserted into the cup **22** such that the three-dimensional gasket **26** is disposed between the holder **24** and the containment bottom **38**. The cap portion **52** of the holder **24** is brought into contact with and joined to the continuous upper edge **36** of the cup **22** (FIG. **12**). Any joining or bonding method available to the artisan can be utilized to join the cap portion **52** to the cup **22**, including gluing, fusion or sonic welding, or a detent-and-groove snap fit engagement.

[0043] After assembly, the orientation of the elongate channels **72** of the holder **24** are such that the channel axes **74** are substantially perpendicular to the junction line **46**. The extended passage **84** is in substantial alignment with the through passage **58** and is substantially concentric about the insertion axis **64**. The pedestal portion **48** can extend partway into the extended passage **84**. The perimeter **50** of the top face **49** of the pedestal portion **48** can be normal to the insertion axis **64**, and can also be concentric about the insertion axis **64**.

[0044] The three-dimensional gasket **26** can be compressed between the holder **24** and the inclined surface **44** of the containment bottom such that the lower face **92** of the perimeter structure **88** is in sealing contact with the inclined surface **44** of the containment bottom **38**. The outer face **94** of the front portion **90** of the perimeter structure **88** can also be in sealing contact with the interior of the first side **32** of the cup **22** just above the junction line **46**. In this way, a chamber **150** is defined, bounded by the diaphragm portion **86**, the perimeter structure **88** and the inclined surface **44**. The chamber **150** is accessible via the extended passage **84**.

[0045] Each of the riblets **102** of the three-dimensional gasket **26** extend into the chamber **150** and are in substantial alignment with a corresponding one of the ribs **68** of the planar face **66** of the holder **24** to form a plurality of channel extensions **152**. Engagement of the lower surfaces **104** of the riblets **102** with the inclined surface **44** creates a plurality of pockets **154** (partially depicted in FIG. **7**) that are proximate the junction line **46**. In one embodiment, the detection end **112** of a given strip **110** accesses the channel extensions **152** and pockets **154** through slots **156** (FIG. **6**) formed proximate the front portion **90** of the perimeter structure **88** of the three-dimensional gasket **26**. The slots **156** can be dimensioned so that the absorbent material **116** of the detection end **112** of

an occupying test strip **110** is in light contact with the interior of the slot **156**. During assembly, the light contact between the test strip **110** and the slot **156** can be enough to hold the test strip **110** in its respective elongate channel **72** until the assembly operation is complete.

[0046] In operation, the sampling swab **140** of the sampling stem **28** is saturated with bodily fluid to be analyzed (e.g., saliva from a person being screened for drug use). The distal portion **134** of the sampling stem **28** is then inserted into the access port **54** and slid through the through passage **58** of the holder **24** and into the extended passage **84** of the three-dimensional gasket **26**. The sampling swab **140** is thereby brought into contact with the pedestal portion **48**. The sampling stem **28** is further pushed into the access port **54** until the male threaded portion **138** of the sampling stem **28** is engageable with the female threaded structure **136** of the access port **54** (or other complementary coupling structures become engageable). The male threaded portion **138** is then threaded into the female threaded structure **136** until the proximal portion **132** of the sampling stem **28** is fully engaged within the access port **54** (e.g., by registration of the knob **135** against an upper periphery of the access port **54**).

[0047] For embodiments utilizing the ratchet-and-pawl mechanism, the threading action (rotation of the sampling stem **130**) causes the pawl **142** to engage with the ratchet notches **144**. This prevents the sampling stem **28** from rotating out of the access port **154**, thus locking the sampling stem **28** in a fixed axial position along the insertion axis **64**.

[0048] The threading action between the proximal portion **132** and the access port **54** also causes the sampling swab **140** to be further extended into the extended passage **84** of the three-dimensional gasket **26**, causing the sampling swab **140** to be compressed against the pedestal portion **48** (when present) or, more generally, against the containment bottom **38**. The securing structure **51** secures the sampling swab **140**, thereby inhibiting the swab **140** from sliding laterally relative to the insertion axis **64**. By securing the sampling swab **140**, more efficient extraction of bodily fluid from the sampling swab **140** is attained during compression.

[0049] The compression acts to wring the bodily fluid out of the sampling swab **140**, which then flows over the pedestal portion **48**, down the inclined surface **44** and along saliva pathways **160** (FIG. **12**) to pool along the inner face **96** of the perimeter structure **88** proximate the lower most portion of the cavity **42**, i.e., into the pockets **154** located proximate the junction line **46**). The pockets **154** that are nearest the center of the three-dimensional gasket **26** (i.e., more proximate the insertion axis **64**) will tend to receive more fluid initially than the pockets **154** located near the extreme corners of the three-dimensional gasket **26**. The riblets **102** can be dimensioned so that the pockets **154** accept only an amount needed to adequately activate the corresponding test strip **110**. Accordingly, as the bodily fluid flows into the nearest of the pockets **154**, these pockets will overflow and cascade over the riblets **102** as the pockets **154** become successively filled. Given an ample quantity of bodily fluid, all of the pockets **154** should receive sufficient bodily fluid for analysis. In one embodiment, the riblets **102** are sized and spaced so that a dozen pockets **154** can hold 1 ml of fluid and the sampling swab **140** sized to accept 20 ml of bodily fluid.

[0050] The depicted embodiments present twelve elongate channels **72** for housing twelve test strips **110**; however, configurations having more or less channels **72** can also be implemented, depending on the quantity of fluid available for analysis. Typically, the number of elongate channels **72** ranges from 6 to 15.

[0051] While the cascading action between the pockets **154** can take several seconds, the absorption by the test strips **110** is on the order of minutes. Therefore, the riblets **102** prevent more absorbent or faster absorbing test strips from drawing test fluid away from less absorbent or slower absorbing test strips, thus providing better and more predictable analysis results.

[0052] The insertion of the distal portion **134** of the sampling stem **28** into the extended passage **84** also acts to pressurize the chamber. The three-dimensional gasket **26** can be made of a soft, pliable material, so that the perimeter structure **88** provides a seal with the inclined surface **44**, and a seal is also affected between the collar **141** and the extended passage **84** as the collar **141** of the sampling

stem **28** reaches its final position within the extended passage **84**. Thus, as the distal portion **134** of the sampling stem **28** moves into its final position, air is displaced and pushed into the chamber **150**. The pliable material can also cause the diaphragm portion **86** to flex, thereby causing the air contained by the three-dimensional gasket **26** to become slightly pressurized. While some venting may occur between these sealed surfaces, as well as through the absorbent material **116** deposited in the slots **152**, the pressurization can still assist in the diffusion of the bodily fluid into the absorbent material **116**, thus augmenting the capillary action between the bodily fluid and the test strip **110**.

[0053] All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, can be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive.

[0054] Each feature disclosed in this specification (including any accompanying claims, abstract and drawings) can be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features. For purposes of interpreting the claims for the present invention, it is expressly intended that the provisions of Section 112, sixth paragraph of 35 U.S.C. are not to be invoked unless the specific terms “means for” or “step for” are recited in the subject claim.

Claims

1. A device for testing a liquid sample, the device comprising: a cup defining an interior cavity, the cup having a planar forward wall and a rearward wall; a holder having an insertion portion positioned in the cup and a cap portion; the cap portion of the holder defining an access port, and the insertion portion has a through passage structure extending from a bottom of the cap portion into the interior cavity of the cup, the through passage structure being in register with the access port; the holder further including a female threaded structure proximate the access port of the cap portion; a swab compression structure; a plurality of test strips arranged in parallel to each other adjacent the planar forward wall of the cup; and a sampling stem insertable through the access port of the cap, the sampling stem having a sampling swab at a distal portion thereof and male threaded portion at a proximal portion thereof, wherein the male threaded portion of the sampling stem engages the female threaded portion of the holder such that rotation of the sampling stem in one rotational direction causes axial movement of the sampling stem such that the sampling swab will be compressed against the swab compression structure.
2. A device for testing a liquid sample as set forth in claim 1, wherein the holder and sampling stem together comprise a ratchet and pawl mechanism operative to lock the sampling stem against backward rotation relative to the holder.
3. A device for testing a liquid sample as set forth in claim 2, wherein the ratchet and pawl mechanism comprises a plurality of ratchet elements on the holder and at least one pawl on the sampling stem.
4. A device for testing a liquid sample as set forth in claim 2, wherein the sampling stem comprises a knob located at the proximal portion thereof.
5. A device for testing a liquid sample as set forth in claim 4, wherein the knob has a knurled outer surface for gripping by a user.
6. A device for testing a liquid sample as set forth in claim 4, wherein the ratchet and pawl mechanism comprises at least one pawl adjacent to the knob.
7. A device for testing a liquid sample as set forth in claim 6, wherein the at least one pawl is situated between the knob and the male threaded portion along an axis of the sampling stem.
8. A device for testing a liquid sample as set forth in claim 6, wherein the distal portion of the

sampling stem comprises a collar at the distal portion thereof that provides a seal with an inner surface of the through passage structure.

9. A device for testing a liquid sample as set forth in claim 1, wherein the cup comprises a base portion with a lower edge that registers the cup in an upright position.

10. A device for testing a liquid sample as set forth in claim 1, wherein the rearward wall is curved.

11. A device for testing a liquid sample as set forth in claim 1, wherein the swab compression structure comprises a pedestal.
